Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which is **true** about asynchronous label propagation's **stability**? 1 / 1 point

- (•) Different runs can yield **slightly different** community structures due to the **random update *order***
- ( ) It never converges to a stable label assignment
- ( ) It always produces the same partition on every run
- ( ) It uses tie-breaking based on highest betweenness

Nice work

Correct! The algorithm is sensitive to the sequence of node updates.

2. Which statement is **true** about **agglomerative** clustering? 1 / 1 point

- (•) It **begins** with each node as its own community and **merges** them iteratively
- ( ) It depends on repeated calculations of edge betweenness
- ( ) It oscillates labels between nodes until convergence
- ( ) It divides a large community into smaller groups based on edge removal

Nice work

Correct! This bottom-up merging defines agglomerative clustering.

3. Which **assumptions** about community formation does agglomerative clustering rely on? (Select **all** correct answers.) 1 / 1 point

- [ ] High-betweenness edges must be removed first
- [x] **Local merges** can gradually form large, cohesive communities

Nice work

Correct. Agglomerative methods build communities from smaller clusters.

- [ ] Each node has a random label assigned at the start
- [x] **Modularity** is used as a guide to judge whether merging is beneficial

Nice work

Correct. Merges occur if they improve modularity.

4. **Prompt** (True/false style with four statements): Which **statement** is **true** about **semi-synchronous** label propagation, as mentioned in the lecture? 1 / 1 point

- ( ) It flips each node's label in a strictly alphabetical sequence with no randomness
- (•) It aims to **avoid** oscillations of purely synchronous updating **while** reducing the randomness of fully asynchronous updating
- ( ) It removes edges according to betweenness before each partial update
- ( ) It never uses node neighbors to decide labels

Nice work

Correct! Semi-synchronous tries to combine the benefits of both approaches.

5. In the Girvan-Newman algorithm, why do we record **all partitions** that form as edges are removed? 1 / 1 point

- ( ) Because removing edges never affects a node's connections
- (•) Because each time we remove an edge and form new connected components, it's a **candidate** partition to later compare using modularity
- ( ) Because the first partition we get is always the best
- ( ) Because each partition has the same modularity score, we pick randomly

Nice work

Correct! We keep track of all possible partitions, then choose the one with the best modularity.

6. In asynchronous label propagation, **why** might different runs of the algorithm yield **different** partitions? 1 / 1 point

- ( ) Because each node always chooses the least frequent label among its neighbors
- ( ) Because modularity is recalculated at each step
- ( ) Because we remove high-betweenness edges in random order
- (•) Because the **update sequence** of nodes is randomized, labels can spread differently

Nice work

Correct! The random order of updates can change the final outcome.

7. When does **agglomerative** hierarchical clustering **stop** merging communities? 1 / 1 point

- ( ) Once each community has the same number of nodes
- ( ) When the betweenness of all edges is zero
- ( ) As soon as we see any decrease in node degrees
- (•) When **no further merge** yields an increase in modularity

Nice work

Correct! The algorithm halts when merges fail to improve modularity.

8. Which assumptions underlie **synchronous** label propagation? (Select **all** correct answers.) 1 / 1 point

- [ ] The algorithm always converges to a single stable partition
- [ ] All nodes update their labels at different times
- [x] Each node observes **only** the current labels of its neighbors, ignoring any "in-progress" updates

Nice work

Correct. Synchronous means no node sees partially updated neighbor labels.

- [x] The entire network updates labels **simultaneously** after each node decides on a new label

Nice work

Correct. Hence the name "synchronous" updating.

9. Which **assumptions** guide the **Girvan-Newman** approach? (Select **all** correct answers.) 1 / 1 point

- [ ] Nodes with the highest degree are always the first to be removed
- [ ] Communities form only when edges are added to connect isolated nodes
- [x] A **bridge edge** is likely to have many shortest paths passing through it

Nice work

Correct. High betweenness usually indicates that the edge connects different regions of the network.

- [x] Splitting the network from **one large piece** down to smaller pieces can reveal communities

Nice work

Correct. Girvan-Newman is a **divisive** or "top-down" algorithm.

10. How does **tie-breaking** occur in **synchronous** label propagation? 1 / 1 point

- (•) Nodes pick among tied neighbor labels **randomly**, but only **after** the round completes, so all updates happen together
- ( ) Nodes choose the label that maximizes modularity across the entire network
- ( ) Nodes always pick the label of the highest-degree neighbor
- ( ) Nodes remove the edge to any neighbor that disagrees with them

Nice work

Correct! During a synchronous step, a node sees the current labels, picks randomly if there's a tie, and then applies that choice simultaneously.

11. Which **assumption** drives **agglomerative hierarchical clustering**? 1 / 1 point

- ( ) We start with all nodes in a single community and remove edges like Girvan-Newman
- (•) Each node begins in its own community, and communities **merge** if they raise modularity
- ( ) Betweenness is the primary factor for deciding how to cluster nodes
- ( ) Every node remains completely isolated throughout the entire process

Nice work

Correct! This bottom-up approach is characteristic of agglomerative clustering.

12. The Girvan-Newman algorithm relies on **removing edges** with the highest betweenness. What is the **key assumption** behind this strategy? 1 / 1 point

- (•) Edges connecting different communities tend to lie on many shortest paths
- ( ) The network is already perfectly partitioned at the start
- ( ) High-degree nodes are the only ones that matter for community detection
- ( ) Removing random edges yields a better partition than removing high-betweenness edges

Nice work

Correct! Girvan-Newman assumes that inter-community edges have higher betweenness.

13. What **assumptions** shape the **asynchronous** label propagation algorithm? (Select **all** correct answers.) 1 / 1 point

- [ ] Edges must be removed to reveal communities
- [x] Updating nodes in a random sequence leads to good communities

Nice work

Correct. The approach is iterative and local, and the random update order is part of the method.

- [ ] Each node's final label must be set by global modularity measures
- [x] A node decides its label using only its current neighbors' labels

Nice work

Correct. The algorithm is purely local; each node looks at its immediate neighbors.

14. Once we **finish** the Girvan-Newman algorithm (i.e., all edges are removed), how do we choose the **best** community structure? 1 / 1 point

- ( ) We always choose the partition with exactly three groups
- (•) We look at the recorded partitions and pick the one with the highest modularity
- ( ) We combine all isolated components back into one big community
- ( ) We select the partition where each node is in its own community

Nice work

Correct! Modularity helps us decide which partition is most meaningful.

15. During each node's update in asynchronous label propagation, how does a node **choose its new label**? 1 / 1 point

- ( ) It picks the same label it started with
- (•) It adopts the **most popular** (most frequent) label among its neighbors; if there's a tie, it randomly chooses among the tied labels.
- ( ) It merges any two communities with the highest modularity score
- ( ) It removes the edge with the highest betweenness to reduce bridging

Nice work

Correct! That's the central rule in asynchronous label propagation.